Practice Set: Polynomial & Horner's Rule

CSE 4108

Structured Programming I Lab — Extra Practice (not official, for self-study)

1. Signal Strength Calculator:

An antenna's signal strength (in dB) at a distance of *x* kilometers follows the formula below. Write a C program that asks the user to enter the value of *x* and outputs the signal strength.

$$\text{strength} = -2x^4 + 3x^3 - x^2 + 5x + 100$$

Hint: *C doesn't have an exponentiation operator, so you'll need to multiply x by itself repeatedly in order to compute the power of x. (For example: x*x*x is x cubed)*

Sample Run:

```
Enter distance x: 2
Signal strength: 98
```

2. Horner's Rule:

Modify the program of the last task so that the signal strength is evaluated using the following formula:

$$\text{strength} = ((((-2x + 3)x - 1)x + 5)x + 100$$

Note that the modified program performs fewer multiplications. This technique for evaluating polynomials is known as *Horner's Rule*.

Sample Run (same x, to verify both programs agree):

```
Enter distance x: 2
Signal strength: 98
```

> Do a pen-and-paper dry run of both formulas before coding, same as Lab 01 Problems 2–3. If your two programs don't give the same output for the same *x*, one of them has a bug — find it before moving on.